Isolated paired Kummer validation replay

run-20261022T1354Z 2026-10-22 UTC

Question. Does the retained, byte-pinned `paired_uncached` executable reproduce the archived uncached candidate and same-process negation-rho records on an isolated copy of all 1,260 public workloads?

Run boundary. Source commit `c63f7ed98cd9`; executable SHA-256 `87d724fd89fa8d22...`. The command was `timeout 3600 env -u KUMMER_TARGET_CAP ./paired_uncached.original`. The first copy-only launch exited 101 before any target because its theta-model input was missing. After adding that pinned input, the complete run exited 0. The host was a shared Intel Xeon 6975P-C with 32 visible logical CPUs; no exclusive CPU reservation was earned.

Observed output. The second launch printed `completed=1260 solved=1260` and `paired_rho=1260`. All seven candidate and seven rho JSONL files match the archived SHA-256 digests. An independent Rust analyzer reproduced all five analysis files byte for byte, including the replay certificate, validator report, 288 price rows, and 45,360 target-price rows. The original study validation gate passes on materialized inputs.

Resource telemetry, this single replay	Recorded value
Wall elapsed	24:50.16
User CPU	1471.89 s
System CPU	18.32 s
Maximum resident memory	287,748 KiB

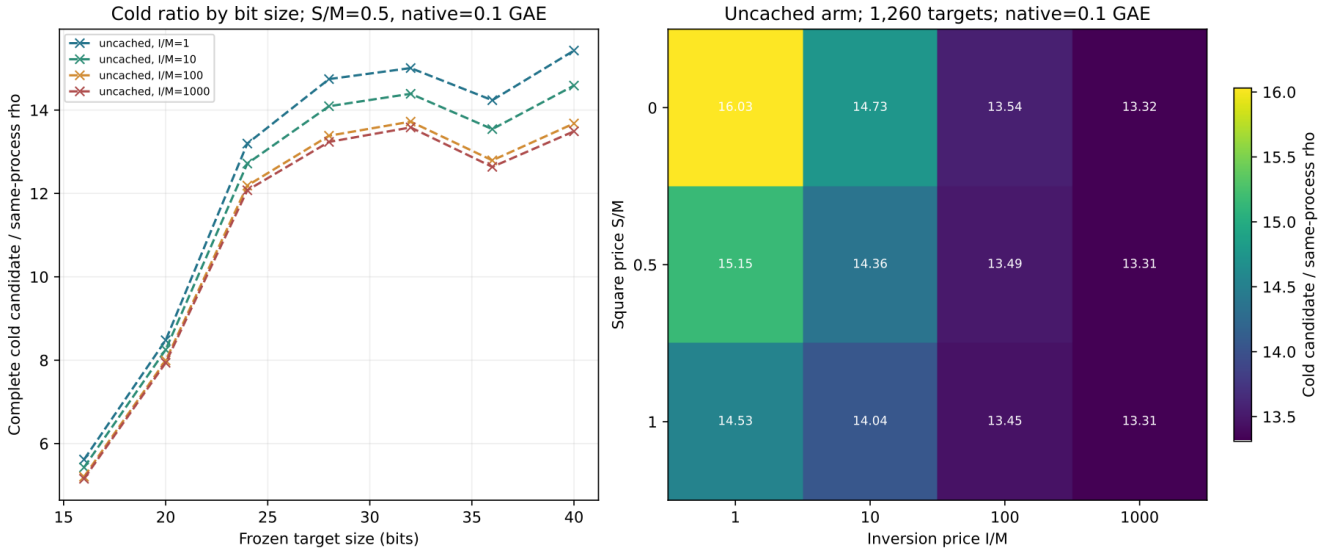
Interpretation. The price graph on page 2 is unchanged because its raw and analyzed inputs reproduced exactly. Across 36 pooled price cells, the modeled cold candidate/rho ratio is 8.959687–55.935424; at $S/M = 0.5$, $I/M = 10$ and 0.1 GAE per native logical operation it is 14.356662. These are operation-cost model ratios, not wall-time ratios. The run does not prove the formulas for arbitrary curves or demonstrate a practical-size attack. The displayed CSV has 45,360 full-cold target-price losses. Its online-only column has 94 individual target-price wins, while all 36 pooled online cells still lose.

Lean proof boundary. The pinned Lean 4.34.1 theorem `pooled_loss_of_target_losses` proves that nonempty aligned natural-number price lists with a strict candidate loss at every target have a strict pooled loss. The repository axiom gate passed. The archived floating-point CSV values and Kummer group formulas are not formal inputs to that theorem, so it does not certify the numerical ratio band.

Evidence. The adjacent `RUN.md` links the input, output, resource and failure logs; the study's `MATCHED_REPORT.md`, `VALIDATION.json` and `plot.py` hold the original comparison, validation gate and graph source.

Archived price graph, reproduced inputs

1,260 public targets; 36 whole-cohort price cells; GAE model units



The candidate's full cold cost includes model, theta and torsion setup charged per target, source transfer, signed anchor, quotient collector and verification. The rho control is the same-process native negation solver on the same workload IDs. The graph is reused from the study because the fresh replay's 14 raw and five analysis files are byte-identical to the archive.